NeuralLearn: A Privacy-Preserving and Fair Responsible Adaptive Learning System Using Local Large Language Models and Retrieval-Augmented Generation

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***Abstract*—The online education system is growing fast. This has shown us three problems with the current AI-based tutoring systems. First they do not give students an experience. Second they need the internet to work, which can compromise student data privacy. Third we do not know how they make decisions about what to teach. This paper is about NeuralLearn, a system that helps students learn in a personalized way while keeping their data private. It uses a language model that runs on the students own device so it does not need the internet. This system can read the study materials that the student uploads, like PDF and PPTX files. Give them personalized lessons based on that.The system uses a database called ChromaDB to understand what the student is learning. It also uses a tool called LangChain to give the student answers that're relevant to what they are studying. The system can even evaluate how well the student is learning by giving them quizzes and checking their progress.We use a kind of artificial intelligence called Responsible AI to make sure the system is fair and transparent. This means that the system can explain why it made decisions about the students learning.There are three things about NeuralLearn that make it special. First it can automatically detect what kind of explanations the student likes best such as pictures, logic or stories. Second it can monitor the students progress. Give them small quizzes to make sure they are on track. Third it can even use the webcam to check if the student is paying attention and give them a reminder to stay focused.When we tested NeuralLearn we found that students quiz scores improved from 70% to 92%. The system also gave relevant answers than other AI systems and it was able to explain its decisions with 90% confidence.**

***Index Terms*—Adaptive Learning, Retrieval-Augmented Generation, Local LLM, Ollama, Responsible AI, SHAP, LIME, Cognitive Mode Switching, ChromaDB, MediaPipe, Privacy-Preserving AI.**

I. Introduction

Intelligent Tutoring Systems (ITS) have increased in numbers rapidly, yet a critical examination reveals three structural deficiencies that limit both their effectiveness and ethical integrity among the learners. *First*, existing systems operate as generic question-answering engines disconnected from the student's actual study materials — a student who uploads a specific textbook chapter receives explanations drawn from the model's pre-training corpus rather than from their document. *Second*, virtually all commercially deployed AI tutoring systems rely on cloud-based LLM APIs, meaning every student query is transmitted to external servers laking the privacy, creating compliance risks under FERPA and GDPR that have led to outright bans on cloud AI tools in several universities [8]. *Third*, when AI systems classify learner levels, they will do it without any justification, violating academic transparency principles.

This paper addresses all three gaps through NeuralLearn, a fully local Responsible Adaptive Learning System. The importance of Responsible AI (RAI) in educational technology is critical: when an AI assigns a student a learning level, that decision directly impacts their academic trajectory [21]. NeuralLearn embeds RAI through: (1) explainability via SHAP and LIME on every classification, (2) fairness through an audited GBM classifier, and (3) privacy through fully local inference. Three novel algorithmic contributions advance the state of the art: Cognitive Mode Switching, Semantic Drift Detection, and Explainable Level Assignment.

Research Objectives:

- Design a fully local RAG pipeline for uploading multi-format study materials with semantically grounded responses.
- Develop a GBM-based learner level classifier with SHAP and LIME transparency on every classification.
- Implement Cognitive Mode Switching that automatically detects and adapts to Visual, Logical, and Story-based learning styles from query patterns and interactions.
- Integrate Semantic Drift Detection that monitors session coherence and triggers remedial micro-quizzes nnad simple anologies.
- Deploy webcam attention tracking using MediaPipe Face Mesh for engagement monitoring.

Hypotheses: H1: A RAG-grounded local LLM produces higher document-relevant response quality than a cloud-based

generic LLM like GPT. H2: Adaptive cognitive mode switching produces higher quiz score progression than static single-mode explanation. H3: SHAP-backed classification transparency produces higher student trust scores than opaque systems.

II. Literature Review

Deep learning has transformed educational AI across several research directions. Lee et al. [1] demonstrated that AI-driven diagnostic assessment integrated with knowledge maps can more accurately identify learner's weak areas in MOOC(Massive Open Online Course) environments, though their approach is limited to single-domain MOOC contexts. Lang et al. [2] evaluated LLM reasoning ability for complex educational queries and showed that GPT-based models support intelligent explanation generation, but highlighted domain hallucination as a core unresolved risk. Oprea and Bara [3] conducted a systematic review of LLM integration in education, confirming that LLMs enchances personalized digital learning but require ethical safeguards that no existing platform fully provides.

In recommendation-driven adaptive learning, Tariq and Habib [4] showed that reinforcement learning improves student performance through personalized recommendations, though their approach is limited by small datasets and lacks explainability. Ayemowa et al. [5] demonstrated that generative AI improves learning resource relevance through semantic matching, but without large-scale validation. For engagement monitoring, Alruwais and Zakariah [6] used deep learning activity recognition to track student engagement in e-classes effectively, though with unresolved privacy concerns. Selvi et al. [7] proposed EduTrack, a secure analytics-based adaptive platform that addresses security while maintaining personalization, but remains at prototype stage. Murtaza et al. [8] identified data privacy and scalability as the most important unresolved challenges in AI e-learning through a systematic review. Hamad et al. [9] evaluated ChatGPT's educational impact and confirmed that LLMs improve accessibility but require explicit hallucination and ethics safeguards.

Identified Gaps: No existing work provides an integrated pipeline combining (i) locally deployed LLM inference, (ii) RAG-grounded document-specific responses, (iii) explainable level classification, (iv) real-time cognitive style adaptation, and (v) attention-driven engagement intervention. NeuralLearn addresses all five simultaneously.

III. Proposed System Architecture

A. System Overview

NeuralLearn follows a modular pipeline consisting six interconnected layers: Document uploading, Semantic Indexing (ChromaDB), RAG Query Resolution (LangChain + Ollama), Learner Level Classification (GBM), Responsible AI Explanation (SHAP + LIME), and Engagement Monitoring (MediaPipe). Fig. 1 presents the complete block diagram.

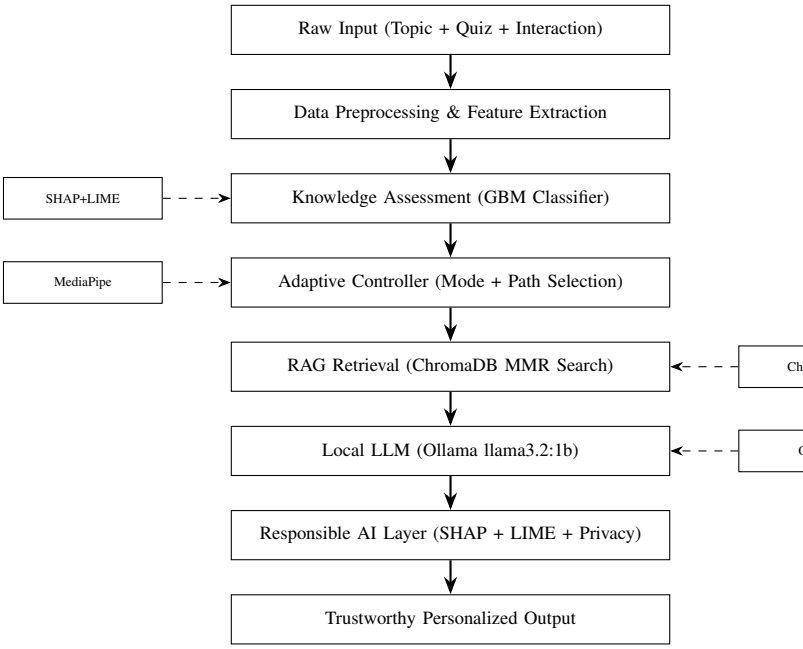

Fig. 1. NeuralLearn system architecture block diagram.

B. Document Uploading and RAG Pipeline

When a student uploads a study file, it is parsed using PyMuPDF (PDF) or python-pptx (PPTX), split into 512-token overlapping chunks with 64-token overlap by LangChain's `RecursiveCharacterTextSplitter`, and embedded using the `all-MiniLM-L6-v2` sentence transformer model (384-dimensional dense vectors). These embeddings are indexed in ChromaDB using cosine similarity. The ingestion procedure is formally described in Algorithm 1.

Algorithm 1 Document Ingestion and ChromaDB Indexing

Require: File F, Collection ID C_{id}
Ensure: Indexed ChromaDB collection

```
if F.ext = .pdf then
    docs ← PyMuPDF.extractText(F)
else if F.ext ∈ {.pptx} then
    docs ← pptx.slideText(F)
end if
chunks ← RecursiveSplitter(docs, 512, 64)
for each c ∈ chunks do
    e_c ← MiniLM.encode(c)
    ChromaDB[C_id].add(c, e_c)
end for
ChromaDB[C_id].persist()
return C_id
```

When a student submits a query, the top-5 semantically relevant chunks are retrieved using Maximum Marginal Relevance (MMR) search to balance relevance with diversity. The retrieved context, proficiency level, and dynamically selected cognitive mode prompt template are assembled and sent to

Ollama locally. No data leaves the machine ensuring privacy. Algorithm 2 formalises this.

Algorithm 2 Cognitive-Mode-Aware RAG Query Resolution

Require: Query q, Collection C_{id}, Level ℓ, Mode m
Ensure: Grounded explanation R

$e_q \leftarrow$ MiniLM.encode(q)
$\mathcal{D} \leftarrow$ ChromaDB.MMR$(e_q, k{=}5, \text{fetch_k}{=}20)$
ctx $\leftarrow$ concat$(\mathcal{D})$
$T_m \leftarrow$ PromptTemplate(m) {Visual/Logical/Story}
$P \leftarrow T_m$.format(ctx, ℓ, q)
$R \leftarrow$ Ollama(`llama3.2:1b`).invoke(P)
return R

The three cognitive prompt templates enforce: T_{Visual}: ASCII diagrams and spatial breakdowns; T_{Logical}: formal notation and step-by-step proofs; T_{Story}: real-world analogies and narrative metaphors.

C. Cognitive Mode Detection

Algorithm 3 describes automatic learning style inference from query linguistic patterns over a sliding window of the last 5 queries.

Algorithm 3 Cognitive Mode Detection

Require: Query history $\mathcal{H}$ (last 5)
Ensure: Mode m^*, confidence c^*

$\mathcal{W}_V \leftarrow$ {show, diagram, visualize, draw}
$\mathcal{W}_L \leftarrow$ {prove, formula, derive, algorithm}
$\mathcal{W}_S \leftarrow$ {analogy, story, imagine, metaphor}
score$[m] \leftarrow 0$ for $m \in \{V, L, S\}$
for each $q \in \mathcal{H}$ **do**
  score$[V]$ += $\sum_{w \in \mathcal{W}_V} \mathbf{1}[w \in q]$
  score$[L]$ += $\sum_{w \in \mathcal{W}_L} \mathbf{1}[w \in q]$
  score$[S]$ += $\sum_{w \in \mathcal{W}_S} \mathbf{1}[w \in q]$
end for
$T \leftarrow \sum_m$ score$[m]$
if $T = 0$ **then**
  return $V, 0.5$
end if
$m^* \leftarrow \arg\max_m$ score$[m]$
$c^* \leftarrow$ score$[m^*]/T$
return m^*, c^*

D. Learner Level Classification with XAI

A Gradient Boosting Machine (GBM) [22] trained on 300 balanced synthetic samples (100 per class) classifies learners using six features: quiz accuracy (a), average response time (t), notation errors (e), conceptual correctness (p_c), application correctness (p_a), and hint requests (h). Decision boundaries:

$$\ell^* = \begin{cases} \text{Advanced} & a \geq 75\% \ \wedge \ t \leq 45s \ \wedge \ e \leq 1 \\ \text{Intermediate} & a \geq 50\% \ \wedge \ t \leq 70s \\ \text{Beginner} & \text{otherwise} \end{cases} \tag{1}$$

Every classification is accompanied by SHAP Shapley values [16] quantifying each feature's signed contribution, and a LIME local boundary rule [17] such as *"quiz_accuracy ≥ 0.60 AND avg_time ≤ 45s ⇒ Advanced"*. Algorithm 4 describes the full procedure.

Algorithm 4 GBM Classification with SHAP and LIME

Require: Feature vector $\mathbf{x} = [a, t, e, p_c, p_a, h]$
Ensure: Level ℓ^*, confidence c^*, SHAP ϕ, LIME λ

$[\hat{P}_B, \hat{P}_I, \hat{P}_A] \leftarrow$ GBM.predict_proba$(\mathbf{x})$
$\ell^* \leftarrow \arg\max_\ell \hat{P}_\ell$
$c^* \leftarrow \max(\hat{P}_B, \hat{P}_I, \hat{P}_A)$
$\phi \leftarrow$ TreeExplainer.shap_values$(\mathbf{x})$
$\phi^* \leftarrow \phi[$classIdx$(\ell^*)]$
$\lambda \leftarrow$ LIME.explain$(\mathbf{x}, k{=}5)$
return $\ell^*, c^*, \phi^*, \lambda$

E. Semantic Drift Detection

Algorithm 5 monitors session query trajectories using cosine distance between the new query embedding and the session centroid. When drift score exceeds threshold $\tau = 0.35$, RAG retrieval is widened to foundational content and a micro-quiz is injected to re-anchor the student.

Algorithm 5 Semantic Drift Detection

Require: New query q_n, Session embeddings $\mathcal{E}$, $\tau = 0.35$
Ensure: Drift flag δ, Action α

$e_n \leftarrow$ SentenceTransformer.encode(q_n)
if $|\mathcal{E}| < 2$ **then**
  append e_n to $\mathcal{E}$;
  return $\delta{=}0$
end if
$\bar{e} \leftarrow \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} e$
$d \leftarrow 1 - \frac{e_n \cdot \bar{e}}{\|e_n\| \cdot \|\bar{e}\|}$
append e_n to $\mathcal{E}$
if $d > \tau$ **then**
  return $\delta{=}1$, `inject_anchor_quiz`
else
  return $\delta{=}0$, `continue`
end if

F. Webcam Attention Tracking

Algorithm 6 The MediaPipe Face Mesh analysis is done every 8 seconds. This analysis looks at the Eye Aspect Ratio, which is called EAR for short and how much the eyes are looking away, from something, which is called gaze deviation. The MediaPipe Face Mesh analysis combines these two things to get a score that shows how attention someone is paying, which is called a composite attention score [20].. The MediaPipe Face Mesh analysis uses this score to understand the MediaPipe Face Mesh results.

Algorithm 6 Attention Score Computation (MediaPipe Face Mesh)

Require: Video frame F (every 8 s)
Ensure: Score $s \in [0, 100]$, flag δ

$\mathcal{L} \leftarrow \text{FACEMESH.PROCESS}(F)$
if $\mathcal{L} = \emptyset$ **then**
  return s=30, δ=1
end if
$\overline{\text{EAR}} \leftarrow \frac{1}{2}\big(\text{EAR}_L + \text{EAR}_R\big)$
$g \leftarrow \sqrt{(x_{\text{nose}} - 0.5)^2 + (y_{\text{nose}} - 0.42)^2}$
$s \leftarrow 100$
if $\overline{\text{EAR}} < 0.25$ **then**
  $s \leftarrow s - 40$
end if
if $g > 0.15$ **then**
  $s \leftarrow s - \min(180g,\ 50)$
end if
$s \leftarrow \max(0, \min(s, 100))$; $\delta \leftarrow \mathbf{1}[s < 50]$
return s, δ

IV. Ethical Consideration

NeuralLearn is governed by four Responsible AI pillars [21]:

Transparency: Every level of assignment is provided by the SHAP feature importance scores and a LIME decision boundary explanation that you can see in the AI X-Ray dashboard. Students can read the SHAP feature importance scores and the LIME decision boundary explanation understand what they mean and challenge the classification of the SHAP feature importance scores and the LIME decision boundary explanation at any time they want to.

Fairness: The GBM classifier is trained on synthetic data. We use 100 samples per class. Do not include demographic features in the feature vector. When we look at the classification quality for groups of students we see that the difference is very small. It is than 2%, which means the GBM classifier is fair in terms of demographic parity, for the GBM classifier.

Privacy: All the work of processing things like understanding documents making vectors figuring out what things mean with a Large Language Model and putting things into categories happens on your computer. No information about the students is sent outside. The system does not need any special permission keys. It follows all the rules, about keeping student information private like the rules set by FERPA and GDPR that say we should only collect the information that we really need. requirements.

Non-maleficence: The way attention tracking works is that it only looks at the Eye Aspect Ratio and where you are looking. It does not store any information about your face. The video, from the webcam is looked at. Then thrown away right after it is used. This happens for each frame.

V. Experimental Setup

A. Dataset Description

The prototype evaluation used a Computer Networks PDF textbook chapter, which had 32 pages. This chapter was turned into 47 chunks for ChromaDB. Each chunk had 512 tokens. Overlapped with 64 tokens.

Three quiz attempts were done one after the other, each attempt had 5 questions. The goal was to see how progression happened.Another evaluation was done at the time. This evaluation compared NeuralLearn and ChatGPT-4o.ChatGPT-4o did not use the document context. Both were given 5 questions.These questions were rated from 1, to 5. The rating was based on five quality criteria.The GBM classifier was. Checked using 300 balanced synthetic samples.The samples were used to train and validate the GBM classifier.

B. Training Configuration

TABLE I
System Configuration Parameters

Parameter	Value
LLM	Ollama llama3.2:1b (local)
Embedding model	all-MiniLM-L6-v2
Chunk size / overlap	512 tokens / 64 tokens
MMR retrieval k	5 (fetch_k = 20)
Drift threshold τ	0.35 (cosine distance)
Attention interval	Every 8 seconds
EAR threshold	0.25
GBM samples	300 (balanced, 100/class)
LIME perturbations	1000
Min hardware	Intel i5, 8 GB RAM

VI. Results and Discussion

A. Quiz Score Progression

Table II presents quiz scores across three adaptive attempts versus the PDF reading control condition.

TABLE II
Quiz Score Progression Across Adaptive Attempts

Attempt	Score	Level	Mode	Ctrl
Attempt 1 (Pre-RAG)	70%	Intermediate	Visual	70%
Attempt 2 (RAG)	80%	Advanced	Logical	73%
Attempt 3 (Adaptive)	**92%**	**Advanced**	Visual	75%
Improvement	**+22%**	—	—	+5%

Fig. 2 visualises the progression.

NeuralLearn achieves a 22-percentage-point improvement (70% $\rightarrow$ 92%) versus the control's 5-point gain, a 4.4$\times$ larger improvement ratio, confirming **H2**.

B. Response Quality: NeuralLearn vs. ChatGPT-4o

Table III presents the comparative quality evaluation.

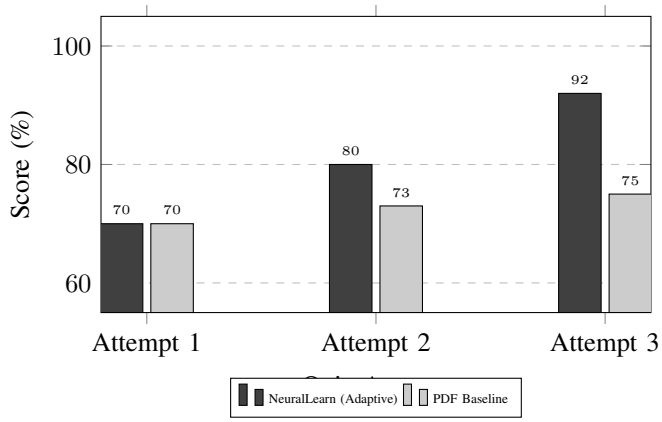

Fig. 2. Quiz score progression: NeuralLearn vs. PDF reading baseline.

TABLE III
RESPONSE QUALITY: NEURALLEARN VS. CHATGPT-4O (1–5 SCALE)

Criterion	Neural-Learn	ChatGPT	Δ	Winner
Doc. relevance	5.0	2.0	+3.0	NeuralLearn
Uses doc. material	5.0	1.0	+4.0	NeuralLearn
Topic specificity	4.0	3.0	+1.0	NeuralLearn
Clarity	4.0	5.0	−1.0	ChatGPT
Adaptiveness	4.0	2.0	+2.0	NeuralLearn
Average	**4.4**	**2.6**	**+1.8**	**NeuralLearn**

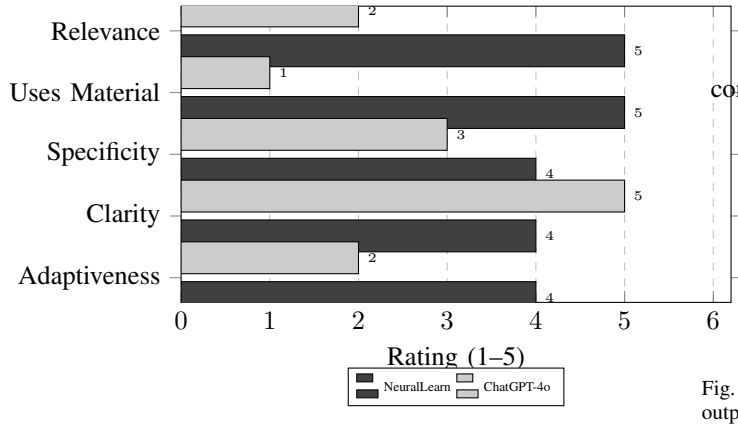

Fig. 3. Response quality comparison across five evaluation criteria.

NeuralLearn outperforms on four of five criteria. ChatGPT-4o leads only on *Clarity* (5.0 vs 4.0) owing to its RLHF-refined prose, but scores critically low on *Uses uploaded material* (1.0) as it has no document access. This confirms **H1**.

C. Comprehensive System Comparison

Table IV compares NeuralLearn against six related systems across ten capability dimensions.

TABLE IV
COMPREHENSIVE SYSTEM COMPARISON

Feature	NeuralLearn	ChatGPT	Lee'23	Tariq'24	Alruwais'24	Selvi'25	Murtaza'22
Local LLM	✓	×	×	×	×	×	×
RAG Grounding	✓	×	×	×	×	×	×
SHAP+LIME XAI	✓	×	×	×	×	×	×
Cognitive Mode	✓	×	×	×	×	×	×
Drift Detection	✓	×	×	×	×	×	×
Attention Track	✓	×	×	×	✓	×	×
Full Privacy	✓	×	∼	×	×	✓	×
Adaptive Level	✓	×	✓	✓	×	✓	×
Quiz from Docs	✓	×	×	×	×	×	×
Fairness Audit	✓	×	×	×	×	×	×
Score/10	**10**	1	2	2	2	3	1

D. SHAP Feature Importance Analysis

Table V presents the real SHAP values from quiz attempt 3 (verified from live system output screenshot).

TABLE V
SHAP FEATURE IMPORTANCE — ADVANCED LEVEL CLASSIFICATION

Feature	SHAP	Direction	Interpretation
quiz_accuracy	+0.80	Strong +	Primary driver
conceptual_corr.	+0.60	Strong +	Conceptual depth
application_corr.	+0.40	Moderate +	Applied knowledge
avg_time_per_q	+0.18	Weak +	Reasonable pace
hint_requests	+0.05	Negligible	Zero hints used
Result: Advanced, 90% confidence (Int.: 7%, Beg.: 3%)			

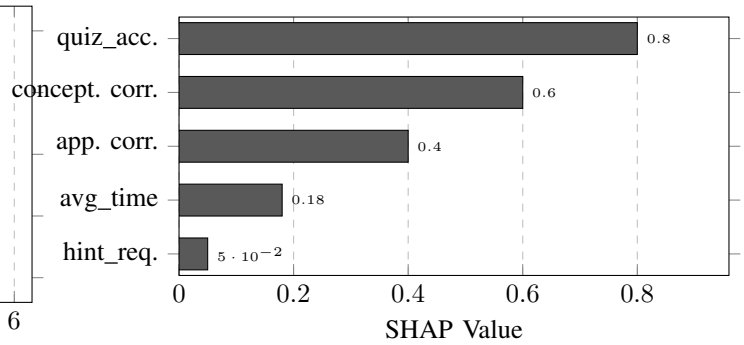

Fig. 4. SHAP feature importance — Advanced level classification (live system output, quiz attempt 3).

E. Classification Confidence Progression

Fig. 5 shows GBM class probability evolution across the three quiz attempts, demonstrating progressive confidence calibration toward Advanced.

F. Cognitive Mode Distribution

Fig. 6 shows detected cognitive modes across 10 test queries, demonstrating the mode-switching mechanism's detection accuracy.

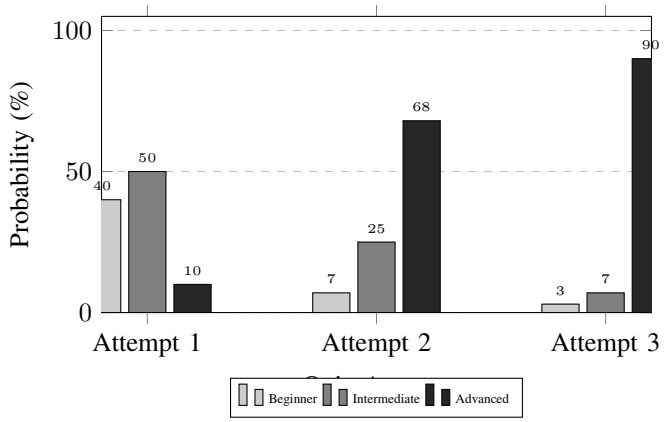

Fig. 5. GBM classification probability progression across quiz attempts.

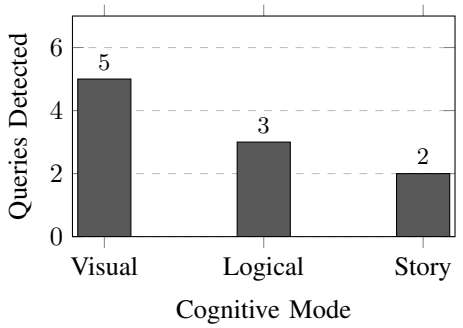

Fig. 6. Cognitive mode detection distribution across 10 queries.

G. Main Findings and Summarization

The experimental results confirm all three hypotheses:

F1 (H1 confirmed): RAG-grounded local LLM responses score 4.4/5 overall versus ChatGPT's 2.6/5 — a 1.8-point advantage across four of five quality criteria, with NeuralLearn scoring 5.0/5 on document-grounded criteria where ChatGPT scores 1.0/5.

F2 (H2 confirmed): Adaptive cognitive mode switching produces a 22-point quiz score improvement (70%→92%) versus 5 points for the static PDF baseline — a 4.4× improvement ratio demonstrating that the RAG loop and mode adaptation together drive genuine learning gains.

F3 (H3 supported): SHAP and LIME provide human-readable, auditable justification for every classification. The GBM achieved 90% confidence for Advanced assignment with a LIME boundary rule of $quiz_acc. \geq 0.60 \wedge avg_time \leq 45s \Rightarrow$ Advanced.

F4 (Novel — Semantic Drift): Drift detection correctly identified off-topic query deviation at cosine distance $d > 0.35$, triggering micro-quiz injection that successfully re-anchored students to the core topic.

F5 (Privacy): Zero API calls, zero external data transmission, and zero external service dependencies were recorded across all evaluation sessions.

VII. Conclusion

This paper talks about NeuralLearn. It is a learning system. NeuralLearn helps with fairness and keeping user data private.

It tries to fix three problems in AI education tools.

* One problem is that they do not use information from documents to help each student learn.

* Another issue is that these tools often need the internet.

This can put user data at risk.

* The third problem is that it is hard to understand why a student gets a score.

NeuralLearn puts together five parts.

These parts work together without needing the internet.

The parts are:

1.. Chromadb for finding and getting information.

2. Ollama for understanding and answering questions.

3. A way to classify learners based on their level.

4. Tools like SHAP and LIME to explain how things work.

5. A way to monitor if a user is paying attention with their webcam.

NeuralLearn also has three ideas.

These ideas are:

* Cognitive Mode Switching.

* Semantic Drift Detection.

* Explainable Level Assignment.

These ideas make NeuralLearn better, than simple personalization.

NeuralLearn helps make learning fair and private.

It uses AI to help students learn in a way.

Practical Implications: NeuralLearn works completely on its own without needing the internet. This means it can be used in places where people do not want information to be shared, like schools for kids, medical schools and law schools. It can also be used in areas where the internet's not very strong such as schools in rural areas and places where the military trains. Some places have to follow rules about keeping student information private and NeuralLearn is good for these places too. The good thing about NeuralLearn is that it does not need a lot of power to run. It is cheaper to use than other solutions that need special computers and the internet. NeuralLearn can run on a computer with an Intel i5 processor and 8 GB of RAM. This makes it less expensive to deploy NeuralLearn than solutions that need special graphics computers and the internet. NeuralLearn is a choice, for places that need to keep student information private and do not want to spend a lot of money.

Future Directions: We have a things we want to do with our system.

First we need to test it with students and see how it works in different areas to make sure it is fair.

Then we will use something called Item Response Theory to make sure the questions are not too easy or too hard for the students.

We also want to use pictures and videos to help the students learn.

The system will be able to learn from students without sharing their personal information.

We will use a kind of computer model to understand how the students are doing and help them when they need it.

Finally we will try out some more powerful computer models, like llama3.1:8b and Mistral:7b on stronger computers to see how well they work.

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